

Stanford AI4ALL

Vision-Language Models for Biomedical Image Analysis

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Outline

1. Artificial Intelligence in Healthcare
2. Exploring MedMNIST
3. Project Outline
    a. What is CLIP?
    b. BiomedCLIP
    c. Radiology & Pathology CLIP Fine-Tuning
4. Challenges/Difficulties
5. Takeaways

Artificial Intelligence in Healthcare



- **Rapid Development:** AI is advancing quickly, offering significant benefits to healthcare.
- **Disease Detection:** AI models can be trained to distinguish between diseased and healthy samples with high accuracy.
- **Broad Applications:** These capabilities have wide-ranging uses across various medical specialties, assisting doctors globally in patient care.
- **Support, Not Replace:** AI is designed to support, not replace, doctors in their roles.
- **Enhancing Patient Care:** As a tool in healthcare, AI aims to improve patient outcomes and overall quality of life.

Project Outline

We split into two groups (Radiology and Pathology). Then we:

- Used a biomedical vision-language model (BiomedCLIP) to classify each medical dataset in a zero-shot fashion.
- Prompt engineered to improve BiomedCLIP's performance on several MedMNIST datasets.
- Fine-tuned a vision encoder from a general vision-language model (CLIP) and surpassed zero-shot and prompt-engineered performance of the biomedical VLM.

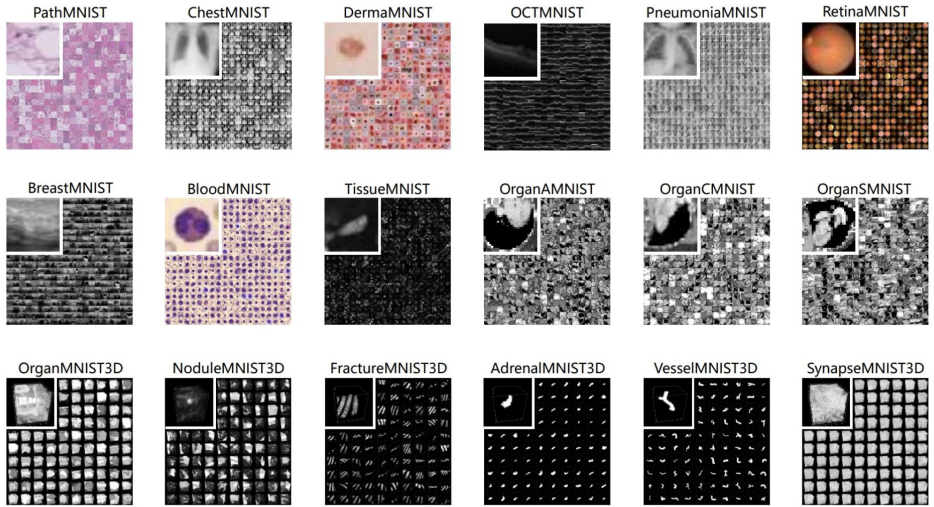






Understanding MedMNIST:

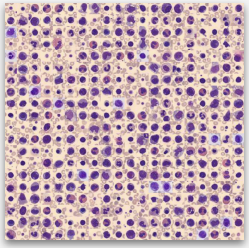
18x Standardized Datasets for 2D and 3D Biomedical Image Classification



Exploring MedMNIST:

- PneumoniaMNIST: Pneumonia Detection [1]
  - **Data Modality:** Chest X-Ray
  - **Task:** Binary Class (2)
  - **Number of Samples:** 5,856
  -
- BloodMNIST: Blood Cell Classification [2]
  - **Data Modality:** Blood Cell Microscope
  - **Task:** Multi-Class (8)
  - **Number of Samples:** 17,092
- DermaMNIST: Lesion Classification [3][4]
  - **Data Modality:** Dermatoscope
  - **Task:** Multi-Class (7)
  - **Number of Samples:** 10,015



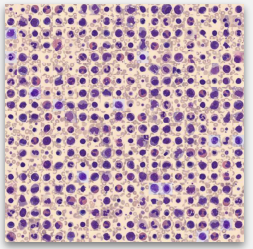




Exploring MedMNIST

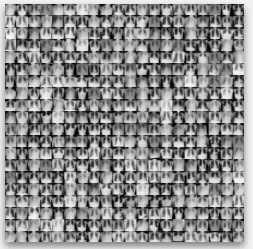
Pathology Team:

- **BloodMNIST:** Blood Cell Classification
  Data Modality: Bright Microscopy
  Task: Multi-Class (8)
  Number of Samples: 17,092



Radiology Team:

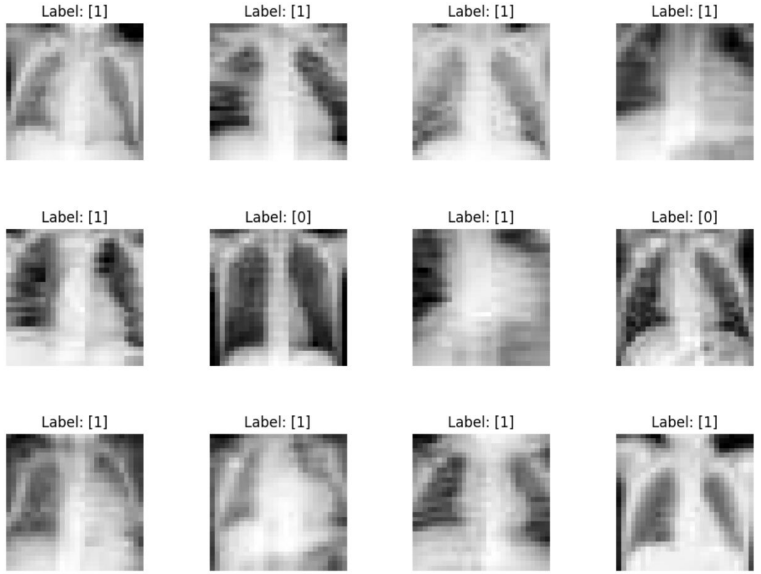
- **ChestMNIST:** Thorax Disease Classifier
  Data Modality: Chest X-Ray
  Task: Multi-Label (14), Binary-Class (2)
  Number of Samples: 112,120



Exploring MedMNIST



PneumoniaMNIST Images before standardization



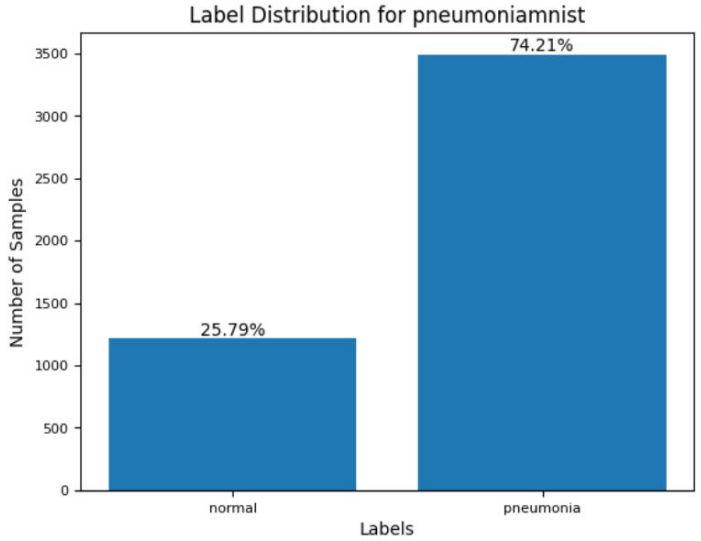
PneumoniaMNIST Images after standardization

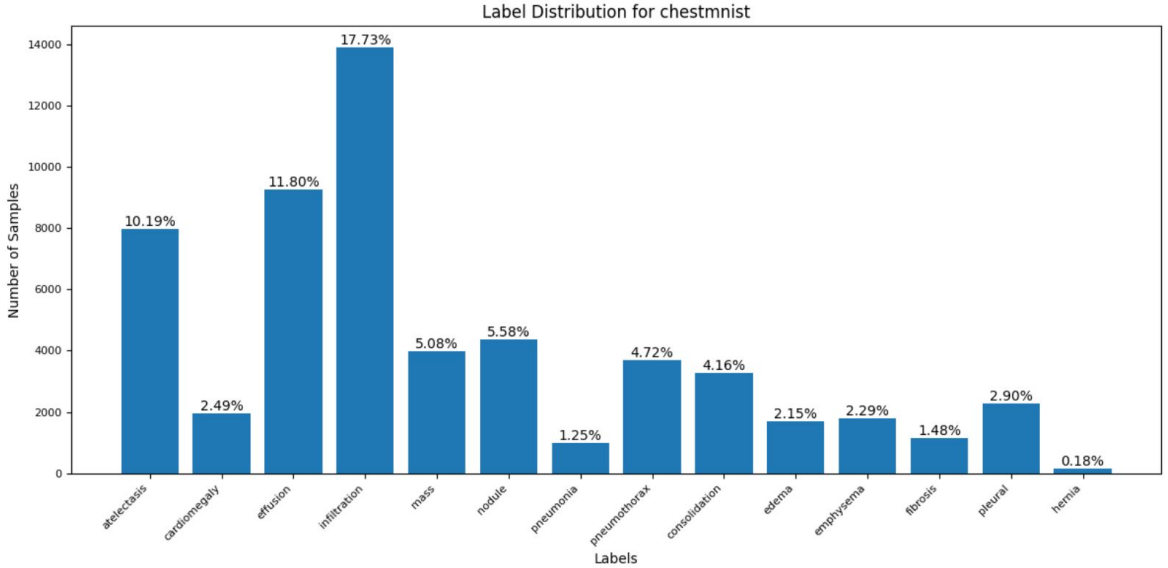
MedMNIST	MIMIC-CXR
• Preprocessed & standardized (224x224) • Lightweight • Easily Accessible • Streamlined for educational purposes	• Stored in DICOM format • 4.7 TB of data • Requires PhysioNet credentialing, training, etc.

Label 0: Normal
Label 1: Pneumonia

Daniel S. Kermany, Michael Goldbaum, et al., "Identifying medical diagnoses and treatable diseases by image-based deep learning," Cell, vol. 172, no. 5, pp. 1122 – 1131.e9, 2018.

MedMNIST Data Distribution (Radiology)



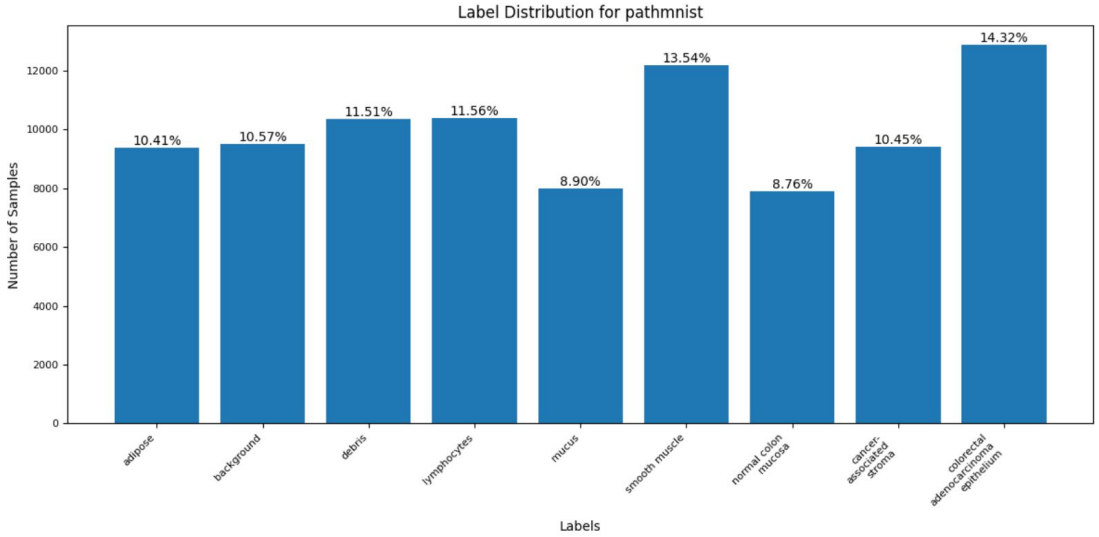


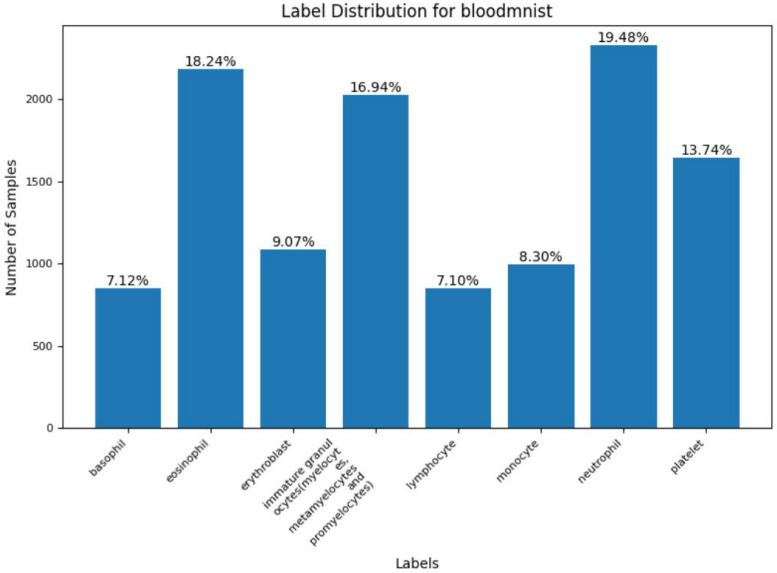
Label 0 (normal): 1214 samples (25.79%)
Label 1 (pneumonia): 3494 samples (74.21%)

Label 0 (atelectasis): 7996 samples (10.19%)
Label 1 (cardiomegaly): 1950 samples (2.49%)
Label 2 (effusion): 9261 samples (11.80%)
Label 3 (infiltration): 13914 samples (17.73%)
Label 4 (mass): 3988 samples (5.08%)
Label 5 (nodule): 4375 samples (5.58%)
Label 6 (pneumonia): 978 samples (1.25%)
Label 7 (pneumothorax): 3705 samples (4.72%)

Label 8 (consolidation): 3263 samples (4.16%)
Label 9 (edema): 1690 samples (2.15%)
Label 10 (emphysema): 1799 samples (2.29%)
Label 11 (fibrosis): 1158 samples (1.48%)
Label 12 (pleural): 2279 samples (2.90%)
Label 13 (hernia): 144 samples (0.18%)

MedMNIST Data Distribution (Pathology)

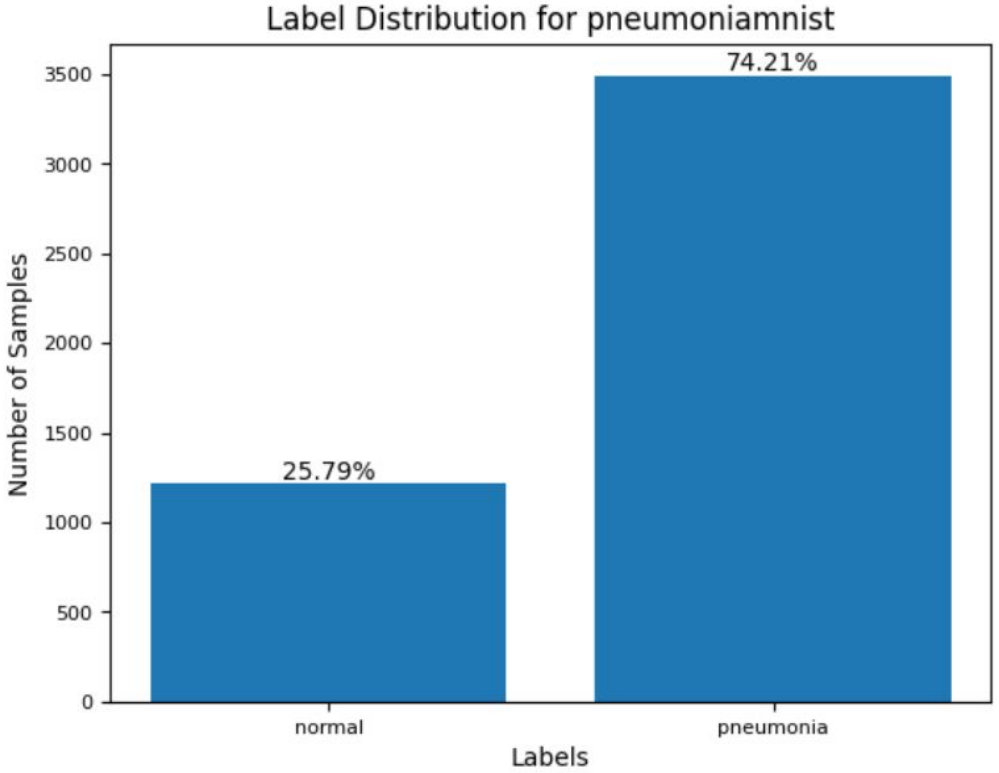




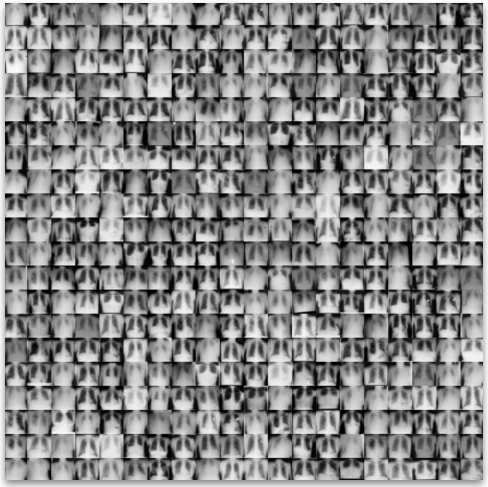
Label 0 (adipose): 9366 samples (10.41%)
Label 1 (background): 9509 samples (10.57%)
Label 2 (debris): 10360 samples (11.51%)
Label 3 (lymphocytes): 10401 samples (11.56%)
Label 4 (mucus): 8006 samples (8.90%)
Label 5 (smooth muscle): 12182 samples (13.54%)
Label 6 (normal colon mucosa): 7886 samples (8.76%)
Label 7 (cancer-associated stroma): 9401 samples (10.45%)
Label 8 (colorectal adenocarcinoma epithelium): 12885 samples (14.32%)

Label 0 (basophil): 852 samples (7.12%)
Label 1 (eosinophil): 2181 samples (18.24%)
Label 2 (erythroblast): 1085 samples (9.07%)
Label 3 (immature granulocytes (myelocytes, metamyelocytes and promyelocytes)): 2026 samples (16.94%)
Label 4 (lymphocyte): 849 samples (7.10%)
Label 5 (monocyte): 993 samples (8.30%)
Label 6 (neutrophil): 2330 samples (19.48%)
Label 7 (platelet): 1643 samples (13.74%)

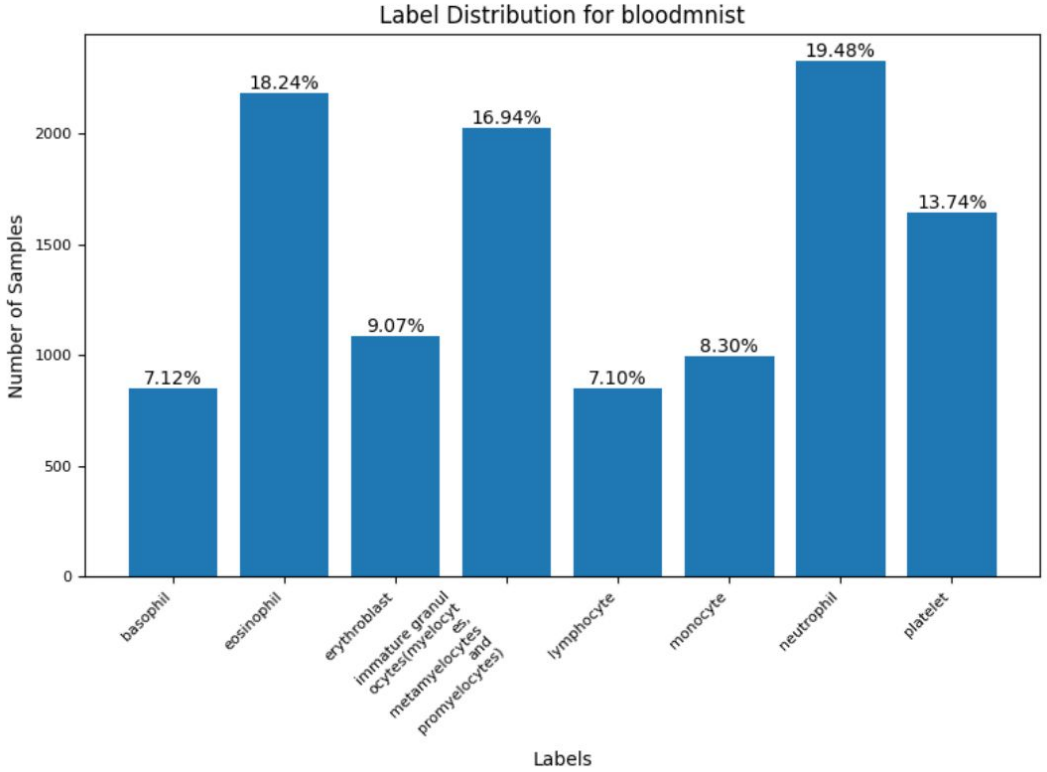
MedMNIST Data Distribution (Radiology)



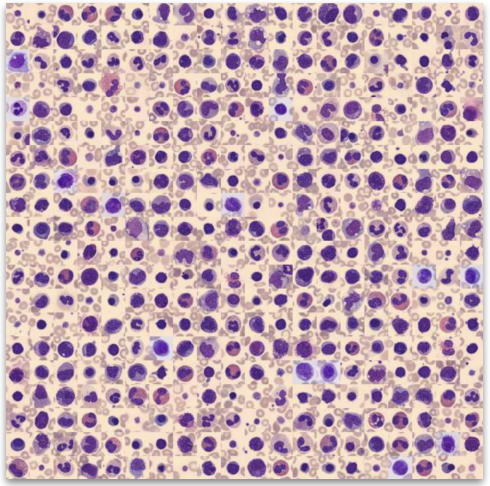
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MedMNIST Data Distribution (Pathology)



Label 0 (basophil): 852 samples (7.12%)
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Understanding Contrastive Image-Language Pretraining (CLIP)



- **Dual-modal:** CLIP is a model developed by OpenAI that can understand and process both text and images.
- **Image Classification:** CLIP can classify images by comparing them to textual descriptions of possible categories via cosine similarity.
- **Text-based Image Search:** CLIP enables searching for images based on textual queries.

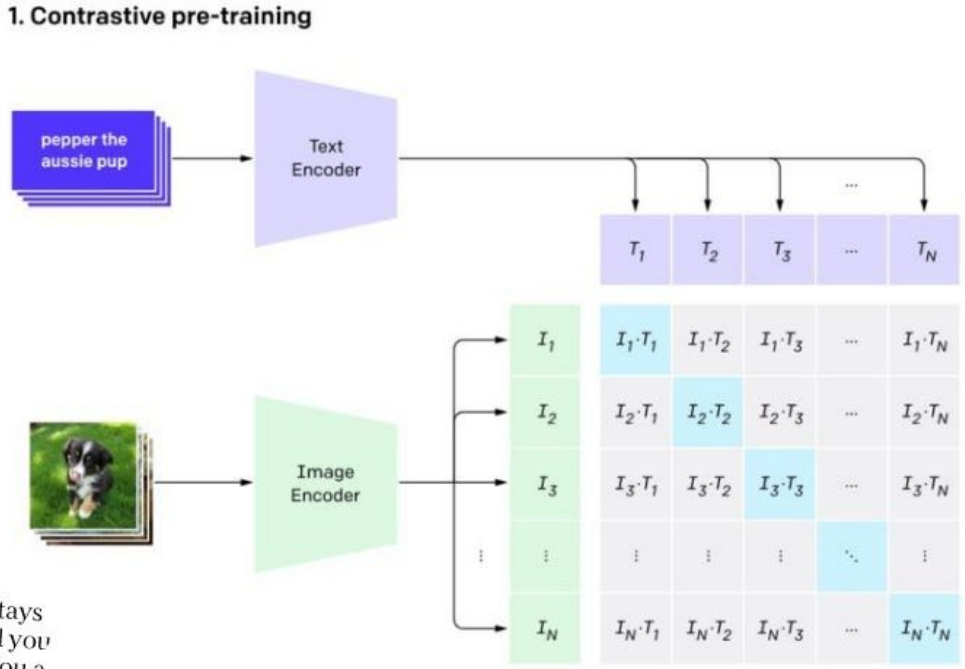
OpenAI CLIP

CONNECTING TEXT AND IMAGES

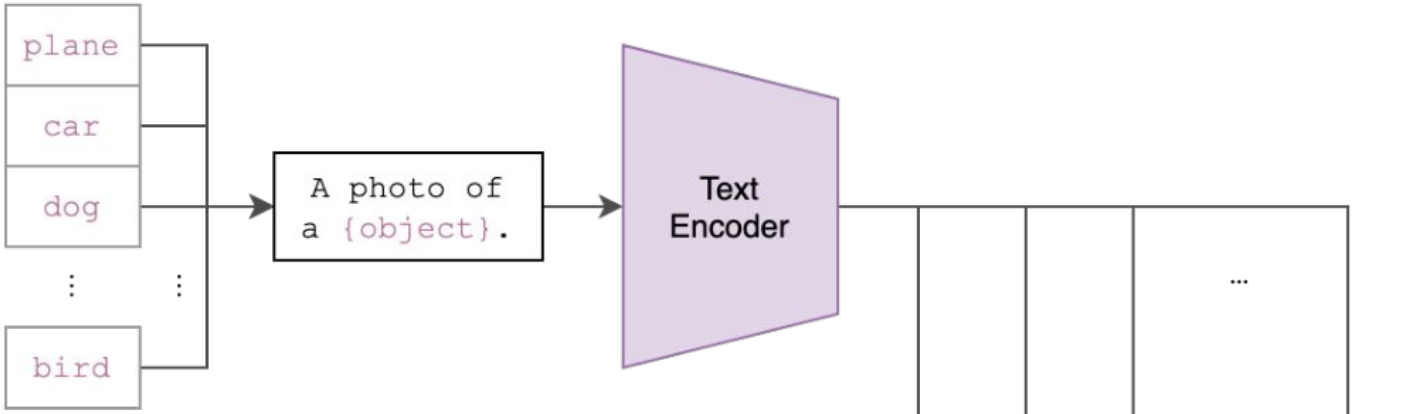




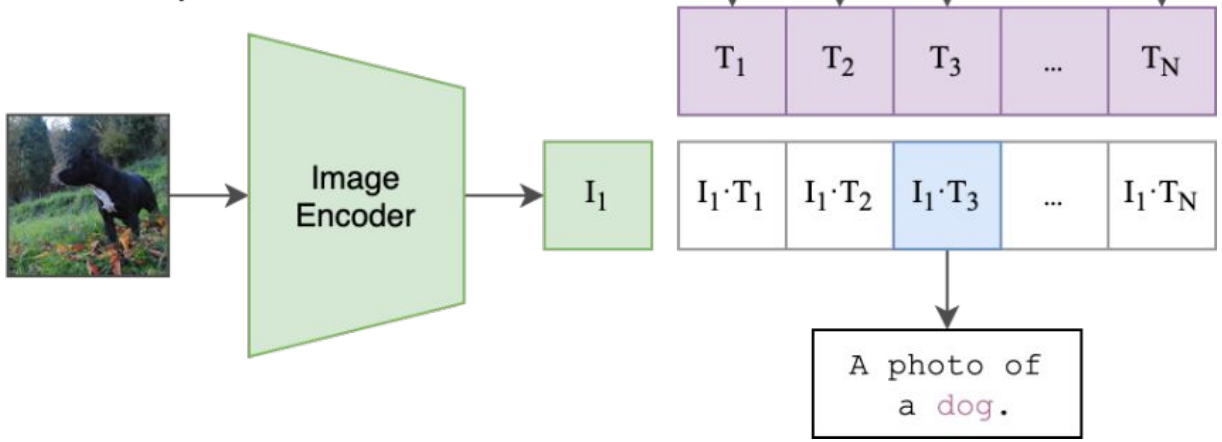


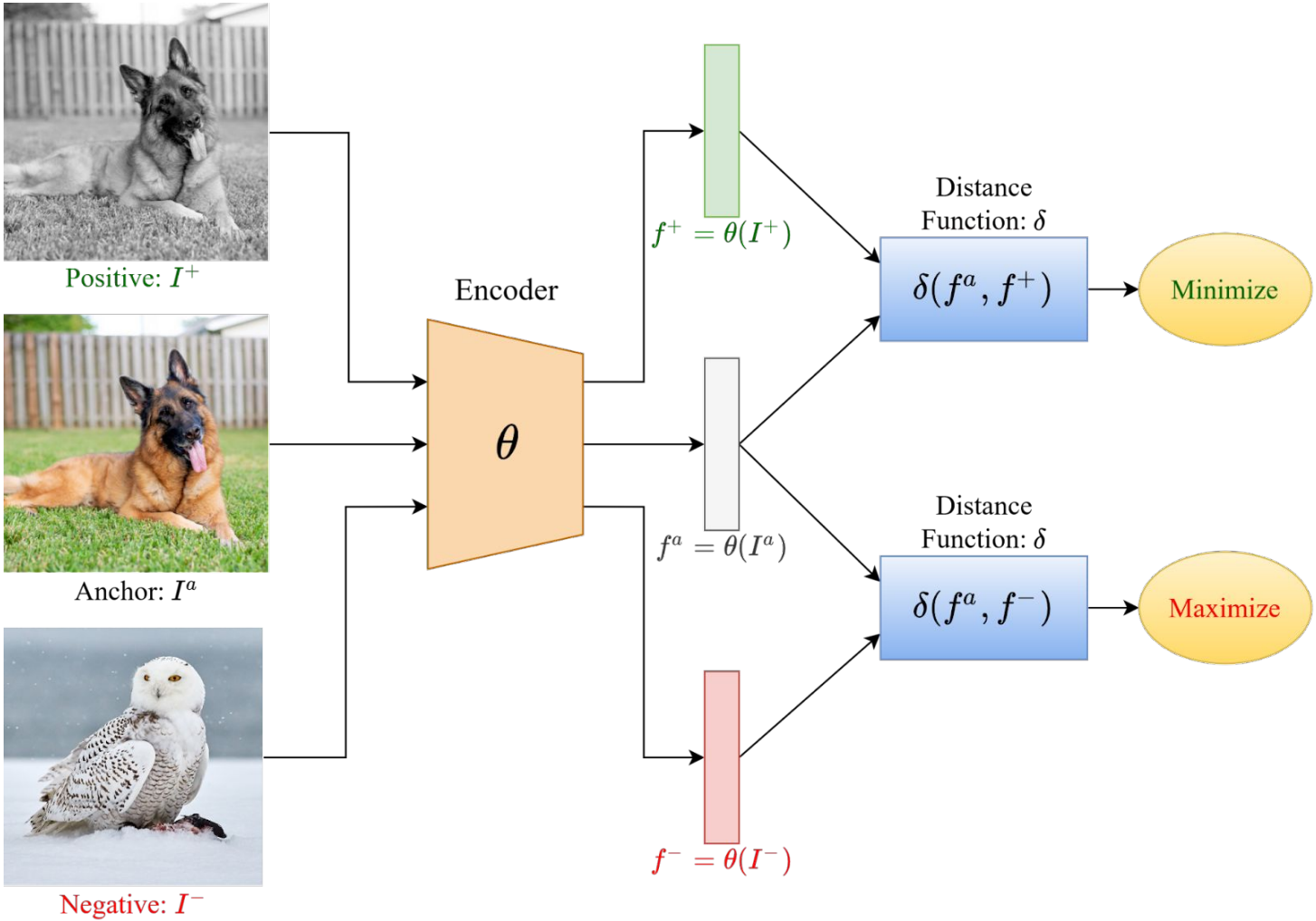


(2) Create dataset classifier from label text

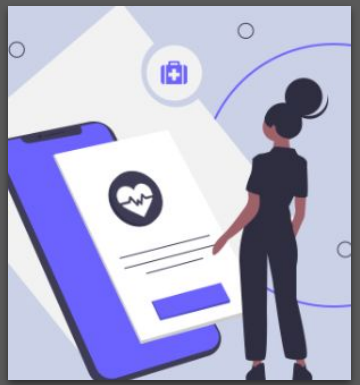


(3) Use for zero-shot prediction




Positive: I^+
Anchor: I^a
Negative: I^-
Encoder
θ
$f^+ = \theta(I^+)$
$f^a = \theta(I^a)$
$f^- = \theta(I^-)$
Distance Function: δ
$\delta(f^a, f^+)$
Minimize
Distance Function: δ
$\delta(f^a, f^-)$
Maximize

BiomedCLIP



- Contrastive language-image model tailored to fit biomedical niches
- PubMedBERT for text encoding
- Enhanced Vision Transformer models for image processing
- Trained on PMC-15M
- Outperformed similar models

BioMed CLIP Prompting

Prompting Engineering

Crafting precise instructions to guide AI systems in producing the desired outputs.

base prompts

```
base_prompts:dict[str,str] =  {"PneumoniaMNIST":"chest X-Ray scan displaying  {class}",
                               "DermaMNIST":    "A skin image siplaying   {class}",
                               "BloodMNIST":    "A micrsocopy image of a blood sample dispaiying  {class}",}
```

tuned prompts

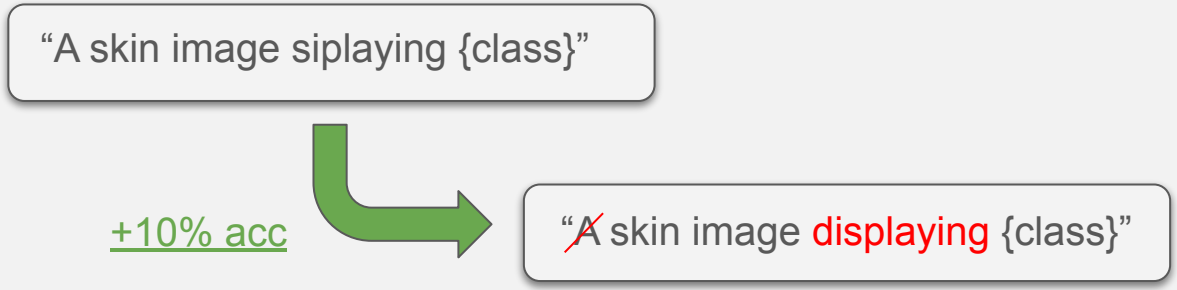
```
base_prompts:dict[str,str] =  {"PneumoniaMNIST":"chest X-Ray image for pneumonia:  {class}",
                               "DermaMNIST":    "skin image for   {class}",
                               "BloodMNIST":    "microscopy blood image: {class}",}
```

	Before (Accuracy)	After (Accuracy)
PneumoniaMNIST	58.8%	87%
DermaMNIST	33.9%	54.4%
BloodMNIST	17.2%	26.1%

Biomed CLIP Prompting

Prompt Engineering

Crafting precise instructions to guide generative AI systems in producing the desired outputs.



	Before (Accuracy)	After (Accuracy)
PneumoniaMNIST	58.8%	87%
DermaMNIST	33.9%	54.4%
BloodMNIST	17.2%	26.1%

Overall Results:

For each of the main 2 datasets cytology and radiology

Have performance (accuracy) of zero-shot, prompt engineering, and fine tuned

	BloodMNIST	ChestMNIST
Zero-shot (BiomedCLIP)		
Prompt engineering (BiomedCLIP)		
Fine-tuned (CLIP)		

Before Fixing Prompts

```
val_datasets["PneumoniaMNIST"]["prompts"] = ["chest X-Ray scan displaying normal",
                                             "chest X-Ray scan displaying pneumonia"]
```

PneumoniaMNIST base prompts

	Accuracy	Balanced Accuracy	F1 Macro	F1 Weighted
Pneumonia MNIST	58.8%	72.2%	58.6%	60%
DermaMNIST	33.9%	23.8%	16.5%	37.4%
BloodMNIST	17.2%	16.8%	10.4%	14.4%

After Fixing Prompts

```
val_datasets["PneumoniaMNIST"]["prompts"] = ["Chest X-Ray scan displaying healthy lung without any abnormalities"
                                             " or areas of opacity or white infiltrates in the lower lobe",
                                             "Chest X-Ray scan displaying signs of infection such as consolidation
                                             " or effusion or areas of opaqueness"]
```

PneumoniaMNIST revised prompts

	Accuracy	Δ	Balanced Accuracy	F1 Macro	F1 Weighted
Pneumonia MNIST	87%	+28.2%	82.3%	82.8%	89.3%
DermaMNIST	54.4%	+20.5%	23.8%	21%	54.5%
BloodMNIST	26.1%	+8.9%	19.2%	12.7%	16.5%

Consider dataset: write prompts in caption style & provide examples

Radiology
CLIP Fine-Tuning

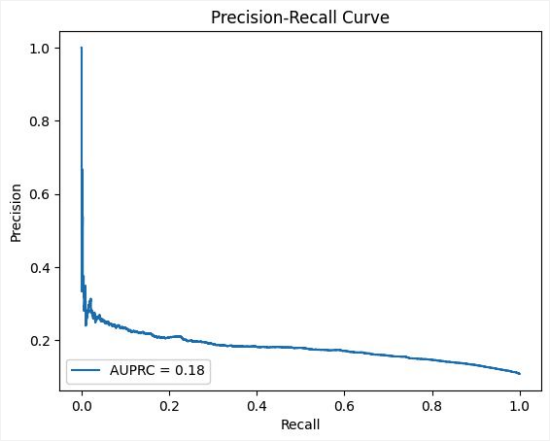
- **OpenAI CLIP:** Used for image-text similarity and zero-shot image classification by utilizing image and text encoders.

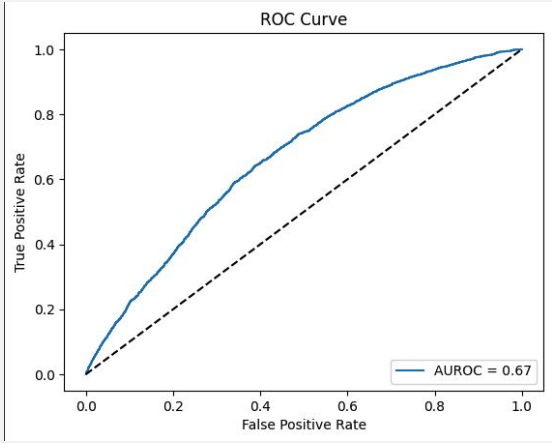
- **Goal:** Fine-tuning the CLIP model on MedMNIST datasets for improved performance on biomedical image classification tasks to better detect and classify diseases.

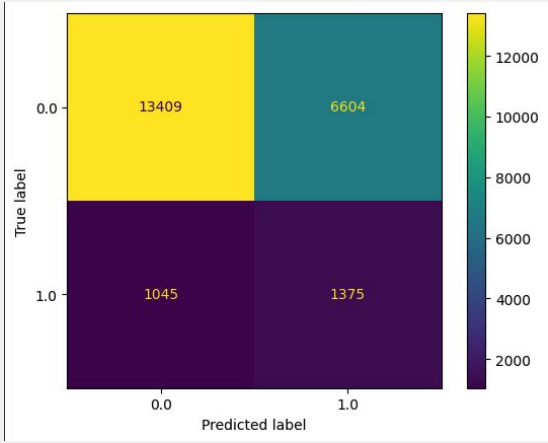
- **Datasets:**
  - PneumoniaMNIST: Determining presence of Pneumonia, Binary Class (2) [1]

  - ChestMNIST: Classifying thoracic diseases, Multi-Label (14) and Binary Class (2) [6]

Radiology CLIP Fine-Tuning Results



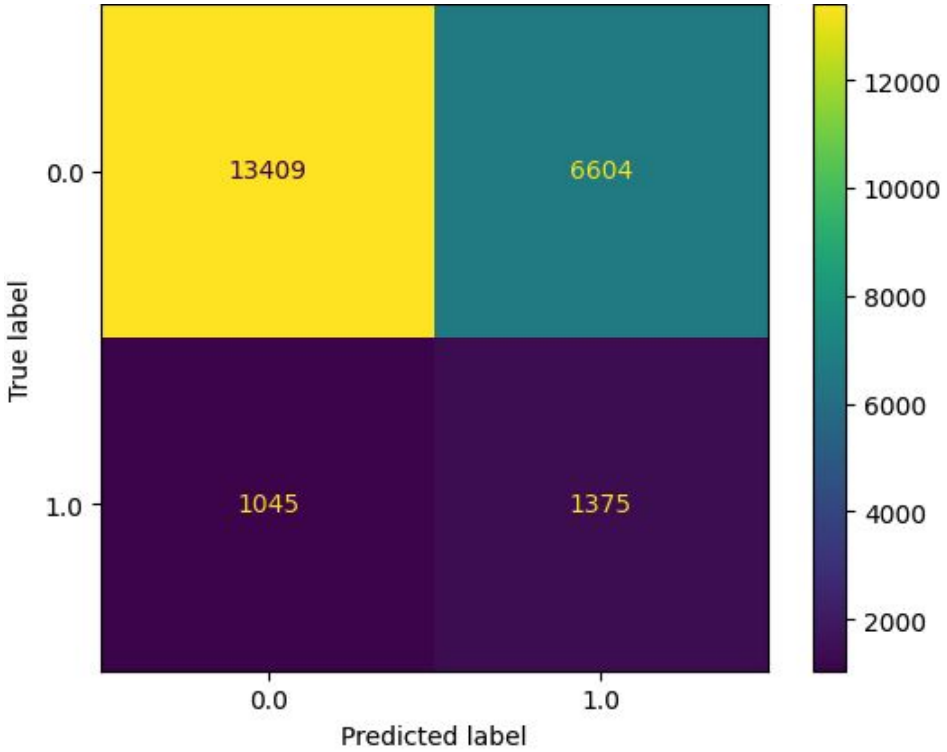


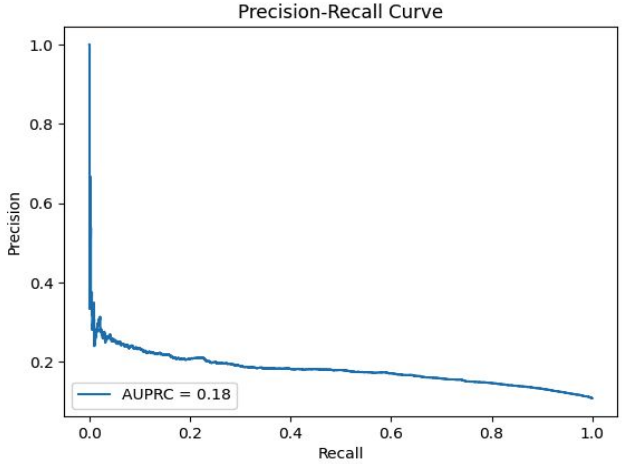


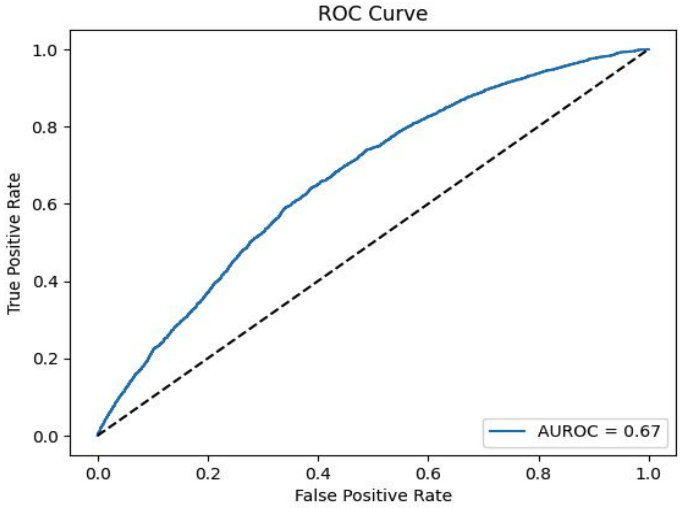
CLIP Fine-Tuning Results



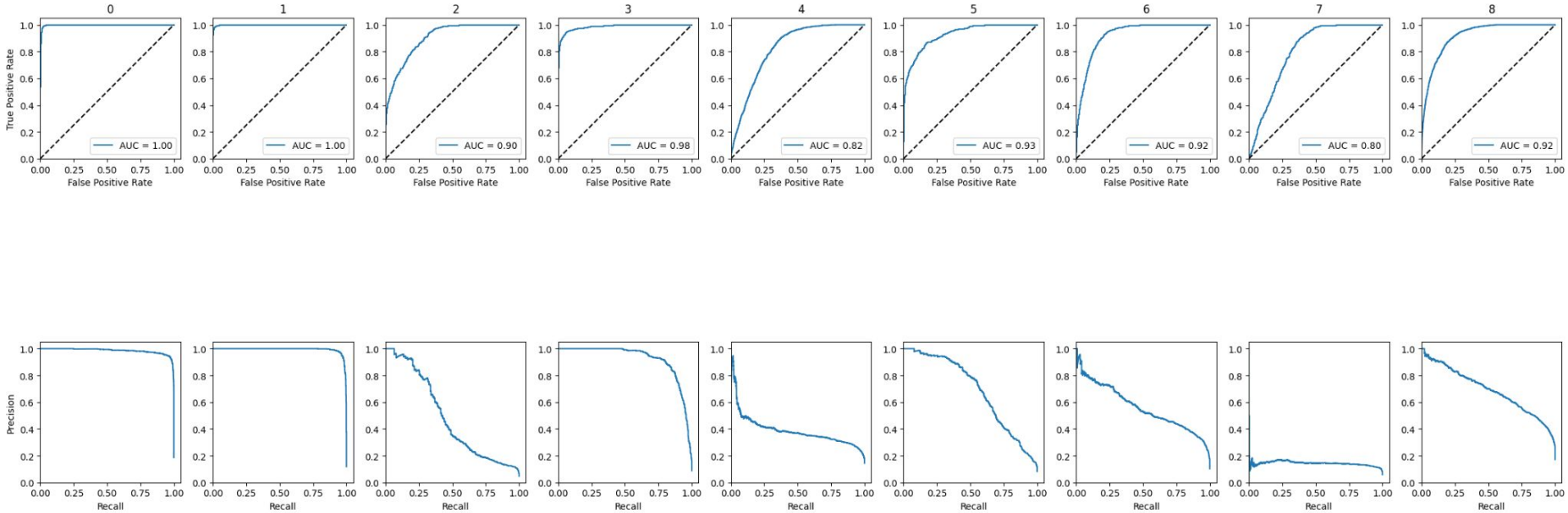
Radiology CLIP Fine-Tuning Results







Pathology CLIP Fine-Tuning Results



Pathology CLIP Fine-Tuning

Confusion Matrix

- Table with four measures
- Used to evaluate AI model

Hyperparameters

- Maximizes model's accuracy
- Different models need different hyperparameters

```
num_classes = cm.shape[0]

for i in range(num_classes):
        TP = cm[i, i]
        FP = cm[:, i].sum() - TP
        FN = cm[i, :].sum() - TP
        TN = cm.sum() - (TP + FP + FN)

        results[i] = {
                'TP': TP,
                'FP': FP,
                'FN': FN,
                'TN': TN
        }

for class_label, metrics in results.items():
        print(f"Class {class_label}:")
        tn, fp, fn, tp = metrics['TP'], metrics['FP'], metrics['FN'], metrics['TN']

        accuracy = (tn + tp) / (tn + tp + fn + fp)
        precision = (tp) / (tp + fp)
        recall = (tp) / (tp + fn)

        print("Accuracy:", accuracy)
        print("Precision:", precision)
        print("Recall:", recall)
```

Challenges/ Difficulties

- **Running out of compute units**
  Google Colab requires compute units to run, and sometimes accounts max out

- **Debugging**
  Errors in syntax, problems with the datasets, and learning how to work with creating the AI model

- **Imbalanced datasets**
  Datasets had different numbers of classes, and some, like PneumoniaMNIST, had mostly images with pneumonia.

- **Varying accuracies and model fit**
  Overfitting (model over trains on training data), underfitting (model under trains on training data), and having poor accuracies with a model are bugs that we had to modify with programming and data

Takeaways

- **Data Quality Impacts the Project**
  High-quality, well-balanced data are ideal for training reliable biomedical AI models. Unfortunately, they are not a reality with the nuances of medical datasets, highlighting the need for rigorous data curation and overcoming challenges with limited datasets
- **Ethical Considerations Are Crucial**
  Ethical frameworks are vital in medical AI development, ensuring patient privacy, preventing biases, and maintaining transparency.
- **Medical AI Continues to Evolve**
  Ongoing research and collaboration are needed to overcome remaining challenges and fully leverage AI's potential in healthcare.

References

[1] Daniel S. Kermany, Michael Goldbaum, et al., "Identifying medical diagnoses and treatable diseases by image-based deep learning," Cell, vol. 172, no. 5, pp. 1122 – 1131.e9, 2018.
[2] Andrea Acevedo, Anna Merino, et al., "A dataset of microscopic peripheral blood cell images for development of automatic recognition systems," Data in Brief, vol. 30, pp. 105474, 2020.
[3] Noel Codella, Veronica Rotemberg, et al., “Skin Lesion Analysis Toward Melanoma Detection 2018: A Challenge Hosted by the International Skin Imaging Collaboration (ISIC)”, 2018, arXiv:1902.03368.Philipp Tschandl, Cliff Rosendahl, et al., "The ham10000 dataset, a large collection of multisource dermatoscopic images of common pigmented skin lesions," Scientific data, vol. 5, pp. 180161, 2018.
[4] Noel Codella, Veronica Rotemberg, et al., “Skin Lesion Analysis Toward Melanoma Detection 2018: A Challenge Hosted by the International Skin Imaging Collaboration (ISIC)”, 2018, arXiv:1902.03368.
[5] Jakob Nikolas Kather, Johannes Krisam, et al., "Predicting survival from colorectal cancer histology slides using deep learning: A retrospective multicenter study," PLOS Medicine, vol. 16, no. 1, pp. 1–22, 01 2019.
[6] Xiaosong Wang, Yifan Peng, et al., "Chest x-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases," in CVPR, 2017, pp. 3462–3471.

Q&A:

We:

- Used a biomedical vision-language model (BiomedCLIP) to classify each medical dataset in a zero-shot fashion.
- Prompt engineered to improve BiomedCLIP's performance on several MedMNIST datasets.
- Fine-tuned a vision encoder from a general vision-language model (CLIP) and surpassed zero-shot and prompt-engineered performance of the biomedical VLM.

